

Executive Summary

Cathy would like to digitalise her company, Pampered Pets. In addition, she plans to internationalise the supply chain and automate warehouses worldwide. The main concerns relate to the potential effects on product quality, system availability, and the security of the supply chain.

This assessment is structured using the ISO 31000 and ISO/IEC 27005 frameworks. ISO/IEC 27005 is used to analyse information security risks related to digitalisation, while ISO 31000 provides a broader framework for evaluating business risks such as supply chain disruption. Monte Carlo simulation is then used as a quantitative technique to estimate the probability of selected disruption scenarios under uncertainty.

Digitalisation significantly increases the organisation's attack surface, as more systems, data, and processes become interconnected and accessible through digital platforms. This increased connectivity creates additional entry points for attackers, particularly in environments where security controls are limited. In small organisations, the lack of dedicated cybersecurity resources further increases exposure, as threats may not be detected or mitigated in a timely manner. As a result, digitalisation increases the organisation's exposure to multiple risks, including cybersecurity threats, data integrity and privacy breaches, and third-party or supply chain failures affecting both availability and product quality.

Cybersecurity disruption, such as ransomware, DoS or DDoS attacks, may represent a significant threat. As a small organisation with limited cybersecurity maturity and no formal staff training, the company is particularly exposed to cyber threats. These attacks could significantly impact availability, as system disruption would prevent customers from making purchases and the company from maintaining or updating stock records. For example, a ransomware attack may enable unauthorised access to internal systems, potentially leading to service disruption and affecting system availability. Similarly, a DoS attack could overwhelm the company's servers, temporarily disable online sales platforms and delay fulfilment operations. In addition, reliance on limited IT infrastructure, including a single server and minimal network redundancy, increases the risk of operational disruption.

Data integrity and privacy breaches also represent a significant risk. In a context of digitalisation and increased data handling, combined with limited staff training, the likelihood of human error is significant. Data may be altered, deleted, or incorrectly processed, which may impact product quality. For example, incorrect product information could lead to errors in preparation or delivery, directly affecting customer satisfaction. Similarly, incorrect inventory data may lead to the preparation of the wrong product. In addition, customer information collected through online sales may be exposed if access controls are weak or if staff make mistakes when handling personal data.

Third party and supply chain failures are another key concern. Increased dependency on external suppliers and logistics providers, combined with international operations, increases

the risk of disruption. Delays in the delivery of materials or inconsistencies in supply may negatively affect both product availability and quality. This is especially important for Pampered Pets because the company is known for high quality products and currently benefits from local sourcing. Moving to international suppliers may reduce costs, but it may also reduce direct control over quality, delivery time, and traceability.

To support the risk evaluation, a quantitative approach was used. Monte Carlo simulation was selected because it helps estimate the probability of disruption in situations where exact internal data is limited. This is appropriate for Pampered Pets because the business faces uncertainty related to cyber threats, human error, and supply chain delays. In this model, the probability range for disruptive cyber incidents was set between 20% and 40%. This range was informed by Cisco (2025), which reported that 44% of small businesses experienced at least one cyber incident, but it was adjusted downward because not every incident would necessarily disrupt operations. The probability range for data alteration, deletion, or corruption was set between 2% and 5%, based on the UK Cyber Security Breaches Survey (2025), which reported that 2% of businesses experienced personal data being altered, destroyed, or taken, and 3% experienced software or systems being corrupted or damaged. The probability range for supply chain disruption was set between 40% and 60%, based on NFIB (2026), which found that 59% of small business owners said supply chain disruptions affected their business to some extent. These ranges were then tested repeatedly across 1,000 simulations in Excel to estimate how often disruptive events may affect product quality or system availability.

The results showed a 65.6% probability of at least one adverse outcome. Among the individual risk categories, supply chain disruption was the most likely at 48.7%, followed by cybersecurity disruption at 28.7%, while data corruption was less frequent at 4.4%. These results suggest that supply chain disruption is the most likely individual risk, while cybersecurity disruption remains the most immediate threat to system availability.

Overall, the three risks differ in probability and in their consequences: cybersecurity primarily affects availability, data integrity affects product quality and operational accuracy, and supply chain disruption may affect both simultaneously.

For this assessment, the model was based on simple assumptions linked to the case study. For example, these assumptions included the possible number of cyber incidents per year, the percentage of staff errors affecting records, the probability of late deliveries from international suppliers, and the average duration of service disruption. Even if precise company data is not available, reasonable assumptions from the case study and literature can still support a useful quantitative estimate. The results therefore provide a probability range rather than one exact figure, which is more realistic in a context of uncertainty. However, these estimates remain dependent on external data and case based assumptions rather than company specific historical records.

These risks may significantly impact the business. In particular, the company's reputation may be damaged if it fails to meet customer expectations, potentially leading to a loss of customers and financial losses. This is especially important in this scenario because the

business has attracted very high profile customers, meaning that even one serious incident affecting quality or availability could have a disproportionate reputational impact.

To mitigate these risks, the organisation should implement appropriate cybersecurity controls, including stronger access management, system protection, and basic security configurations. In particular, multi-factor authentication (MFA) should be implemented, as it reduces the likelihood of unauthorised access by adding an additional layer of security. Least privilege access should be applied by restricting access rights to the minimum required. Regular system updates reduce the risk of exploitation of known vulnerabilities. Regular security audits should also be implemented. In addition, staff awareness training is essential to reduce the risk of human error, particularly in a small organisation with no formal cybersecurity training.

To prevent data loss and ensure data integrity, regular data backups should be implemented to reduce the impact of data loss by enabling system recovery, along with appropriate validation processes to ensure data accuracy. Audit logs, restricted editing rights, and checking mechanisms for stock and order information would also help reduce the likelihood that incorrect data affects product quality.

Third party risk management should also be strengthened by monitoring suppliers and ensuring minimum security and reliability standards across the supply chain. In addition, diversification of suppliers is important to reduce dependency on a single source. Supplier reviews, contractual security expectations, and quality control checks should be included to

reduce both supply disruption and the risk of inconsistent materials entering the production chain.

Business continuity and disaster recovery measures are critical to ensure operational resilience and to mitigate disruptions affecting availability and data integrity. These should include backup arrangements, redundancy of key systems, and alternative connectivity solutions, such as failover options, to maintain operations in case of disruption. For example, in the event of a ransomware attack, backups would allow the organisation to restore its systems and resume operations without paying a ransom. Similarly, redundancy of key systems ensures that if one system fails, another can take over, maintaining business operations.

However, the disaster recovery solution must also meet the business requirements. The online shop must remain available 24/7/365, with a changeover window of less than one minute and with no more than one minute of data loss. The organisation would need a more resilient solution based on high availability, automated failover, near real-time replication between environments, and backup arrangements that support recovery beyond the primary failover environment. In other words, the target should be an RTO of less than one minute and an RPO of less than one minute in order to meet Ms O'dour's expectations.

A cloud platform such as Microsoft Azure may be appropriate to host this solution because it provides scalable infrastructure, geographic redundancy, backup options, monitoring tools, and disaster recovery services. A cloud-based disaster recovery architecture hosted on Microsoft Azure, with near real time replication, automated failover, and geographically separate environments, would better support the required RTO and RPO targets. This would support business continuity for the online shop and improve resilience if the physical shop

premises were affected by a disaster. However, such a solution may be costly and operationally complex for a small business, meaning that the required resilience level may exceed what is proportionate unless the commercial value of continuous availability clearly justifies the investment.

Nevertheless, the company should also consider the risk of vendor lock-in. This means becoming too dependent on one provider's services, tools, or formats, making it difficult or costly to move to another provider later. This risk may be reduced by keeping backups in portable formats, documenting recovery procedures clearly, and avoiding unnecessary dependence on highly proprietary services where possible. A multi-supplier approach may also reduce overdependence on one provider where commercially feasible.

In addition, GDPR requirements should be considered because digitalisation and CRM use will increase the volume of customer personal data processed by the business. Appropriate controls should therefore include restricted access to personal data, secure storage, data minimisation, and clear responsibilities for third-party processors. This is important not only for compliance, but also for maintaining customer trust and reducing legal and reputational risk

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Finally, continuous monitoring and regular updates of the risk assessment are required to adapt to evolving threats, particularly in a context of digitalisation and international expansion. For example, regular system monitoring may help detect unusual activity early, reducing the impact of a potential cyberattack. Similarly, regular reassessment of supplier

reliability may help the business identify weaknesses before they seriously affect product quality or availability.

Overall, although the estimated probabilities differ across the identified risks, their potential impact on both product quality and system availability justifies the implementation of appropriate mitigation measures. Cybersecurity risks represent the most immediate threat due to their direct impact on operations, while data integrity and supply chain risks may have broader consequences on product quality and customer satisfaction.

Therefore, the main priorities should be to protect online availability and customer data, maintain data integrity to protect product quality, and strengthen supplier oversight, through a balanced approach combining technical controls, staff training, and resilient operational strategies. The business should proceed with digitalisation, but only with security, resilience, and risk management integrated from the start. Continuous monitoring and regular reassessment of risks will be critical to maintain an effective security posture, especially in a context of ongoing digitalisation and international expansion.

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